MCC Entrance Test, August 2026 — Day 2

Level T4

Test Information. Level T4 — 180 minutes.
6 problems: 2 Easy (10 pts), 2 Medium (15 pts, two parts), 2 Hard (25 pts, three parts); 100 points total.
Areas: Algebra (×2), Combinatorics, Geometry, Number Theory (×2).
Rules: show your work (full, justified arguments earn full credit at this level); no calculators; read all six before choosing where to start; box your final answers.
A past MCC entrance test, free to try: https://nghia71.github.io/readiness

Problem 1 (10 pts). Consider the numbers 11111, 111111, and 122222 (note that $122222 = 11111 + 111111$). Raising these to the 4th power directly produces numbers too large for an ordinary calculator to evaluate exactly. Find, without a calculator,

$$\sqrt{\frac{11111^4 + 111111^4 + 122222^4}{2}}.$$

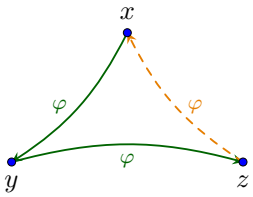
Problem 2 (10 pts). Let x and y be non-negative integers. Find all pairs (x, y) for which $x + 3^y$ and $y + 3^x$ are *consecutive* integers (that is, they differ by exactly 1, in either order).

Problem 3 (15 pts).

Define the function

$$\varphi(r) = r^3 - 4r^2 - 16r + 60.$$

Starting from an integer x, let $y = \varphi(x)$ and $z = \varphi(y)$; say the triple (x, y, z) *closes* if $\varphi(z) = x$ once again.



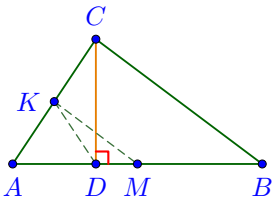
(a) Find all real numbers r with $\varphi(r) = r$. [5 pts]

(b) Determine all integer triples (x, y, z) for which the loop closes as described. [10 pts]

Problem 4 (15 pts). Consider a row of 19 slots, each containing one of two symbol types, labeled 1 and 2. Call such a sequence *balanced* if, among its 16 windows of 4 consecutive slots, no two windows show the same pattern of types.

(a) Prove that a balanced sequence exists. [6 pts]

(b) Prove that in every balanced sequence, the types in the first three slots match, in order, the types in the last three slots. [9 pts]



Problem 5 (25 pts). In triangle ABC with $\angle A \geq \angle B$, point M is the midpoint of side AB, and point D is the foot of the altitude from C to line AB.

(a) Prove that if $\angle A = 2\angle B$, then $AC = 2\,MD$. [10 pts]

(b) Let K be the midpoint of AC. Prove that triangle KDM is isosceles with $KD = DM$ if and only if $\angle A = 2\angle B$. [4 pts]

(c) Prove the converse: if $AC = 2\,MD$, then $\angle A = 2\angle B$. [11 pts]

Problem 6 (25 pts). Let x, y be integers and let p be a prime number satisfying

$$x^2 - 3xy + p^2y^2 = 12p.$$

(a) Show that if $y = 0$, then $p = 3$, and find the two resulting values of x. [5 pts]

(b) Show that if $y \neq 0$, then p must satisfy $4p^2 - 48p - 9 \leq 0$, and conclude that only $p \in \{2, 3, 5, 7, 11\}$ need be checked. [6 pts]

(c) Complete the analysis, and find all triples (x, y, p). [14 pts]